

PSI

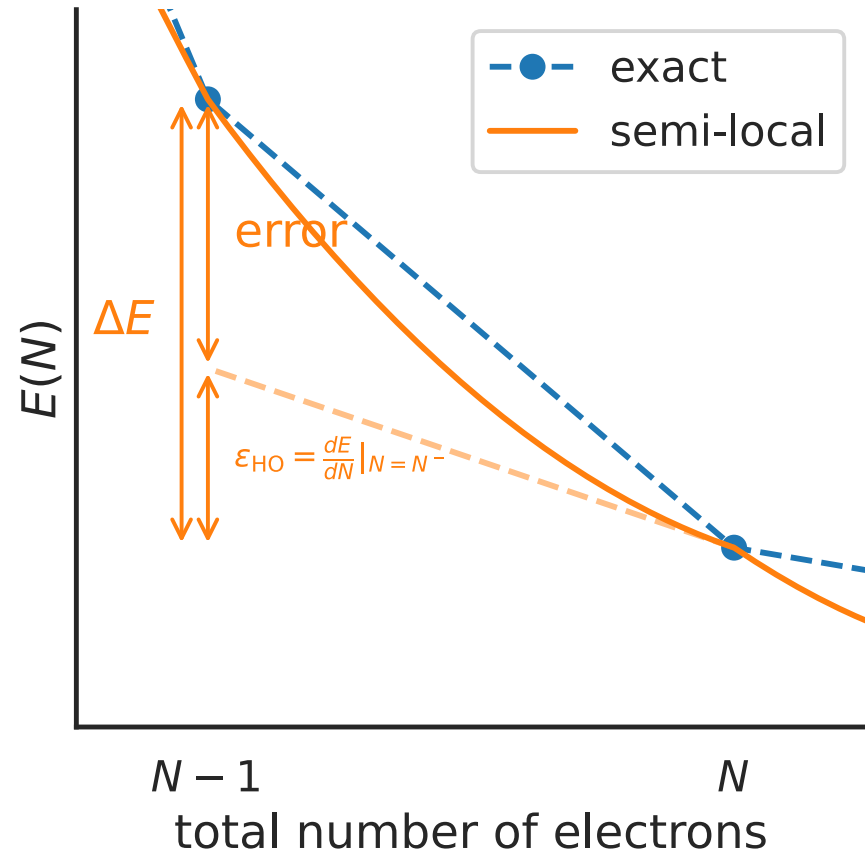
Center for Scientific Computing,
Theory and Data

Koopmans functionals in practice

minimisation, screening coefficients, automation, and more...

Edward Linscott

ICTP–MARVEL College, 27 May 2026



$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

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Differences to semi-local functionals:

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- different flavours

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Differences to semi-local functionals:

- screening
- different flavours
- orbital-density dependence

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Screening

Screening

In Hartree-Fock

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|}$$

C. Li *et al.* *J. Chem. Phys.* **146**, 74107 (2017)

T. Koopmans. *Physica* **1**, 104–113 (1934)

In Hartree-Fock

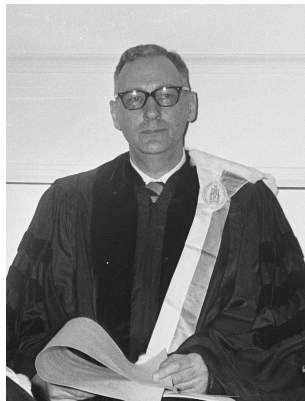
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In Hartree-Fock, the first ionisation energy of a system is equal to the negative of the highest occupied orbital's energy $-\epsilon_{\text{HOMO}}^{\text{HF}}$.

— Koopmans' theorem, 1934

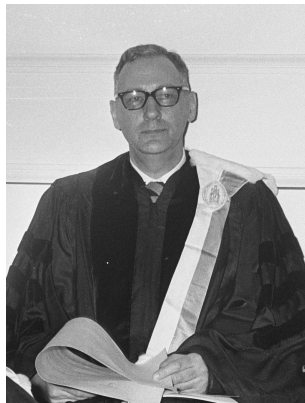
$$\text{IP} = E(N - 1) - E(N) = -\epsilon_{\text{HOMO}}^{\text{HF}}$$

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$$\text{IP} = E(N - 1) - E(N) = -\epsilon_{\text{HOMO}}^{\text{HF}}$$

... exact only if the remaining orbitals do not relax.

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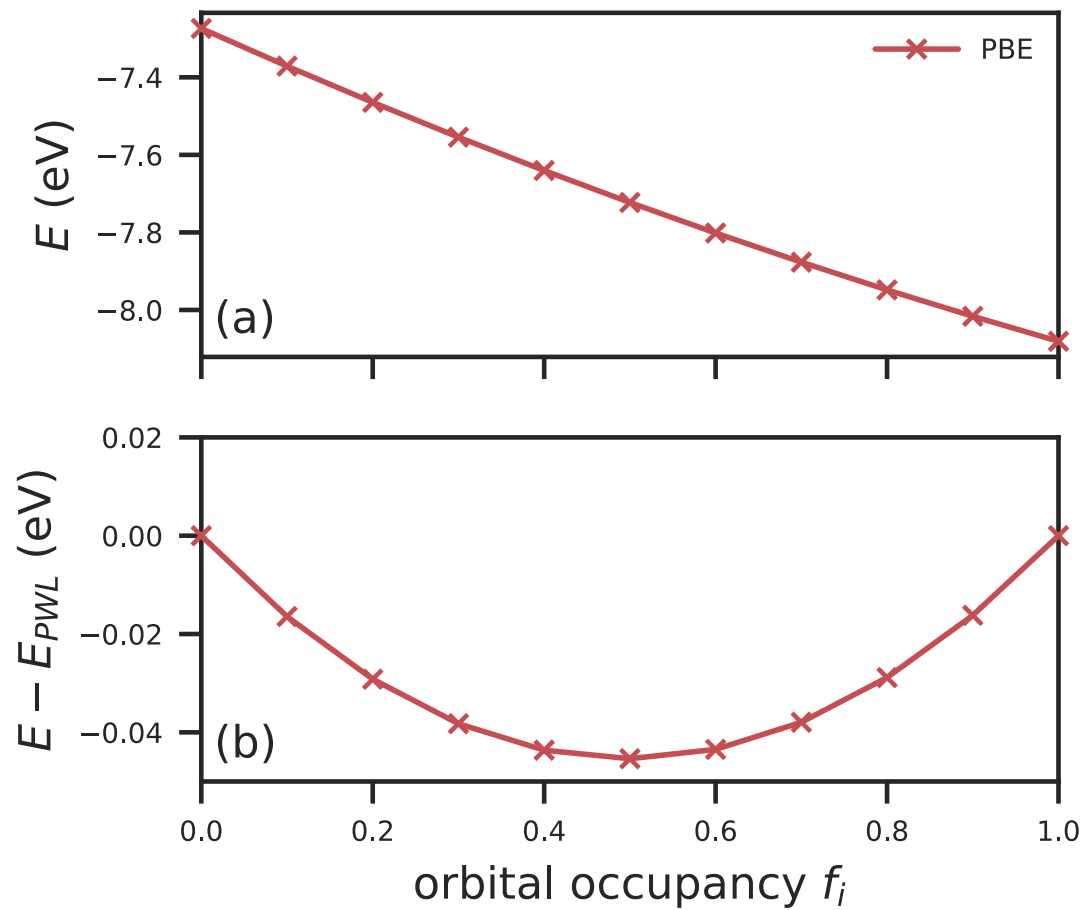
Koopmans functionals

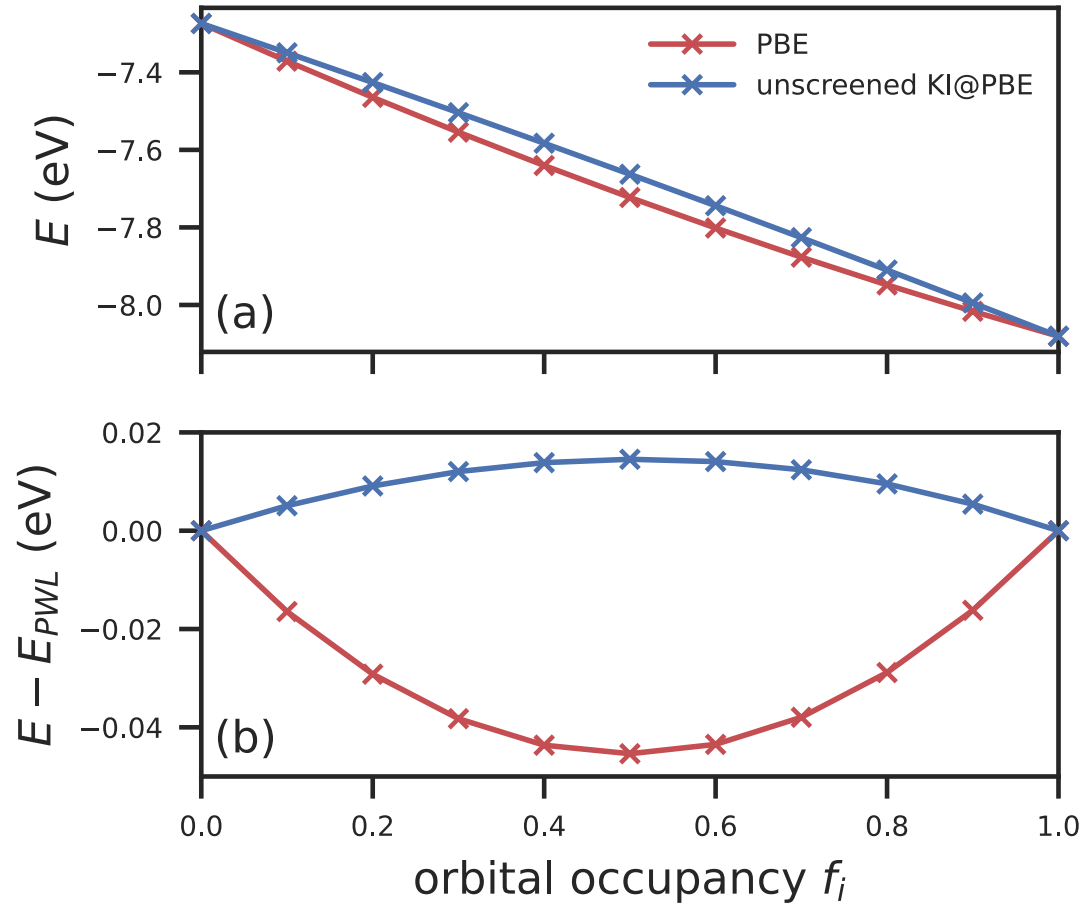
$$E_{\text{KI}}[\{\rho_i\}] = E_{\text{DFT}}[\rho] + \sum_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right)$$

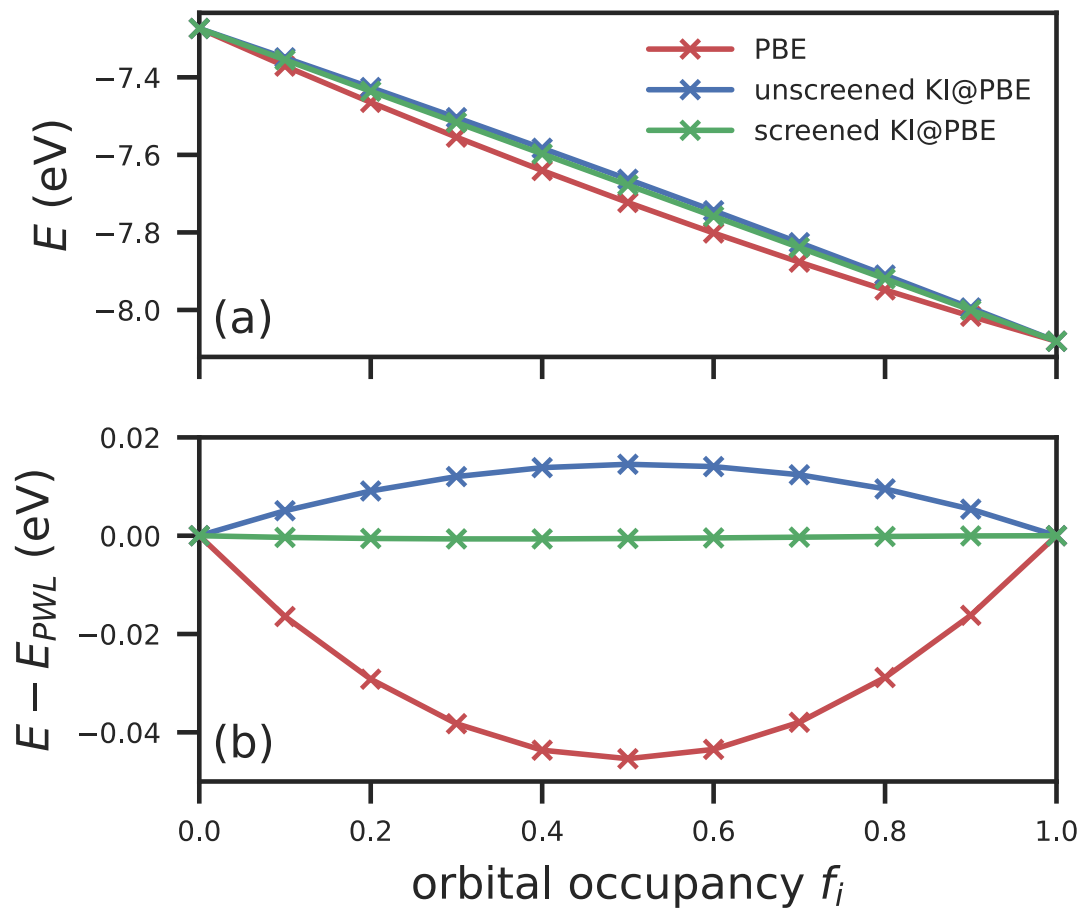
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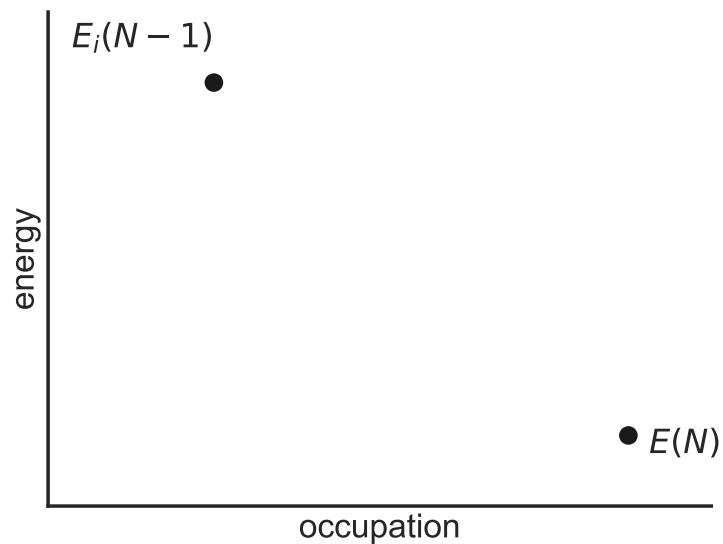
cannot be evaluated directly total energy differences



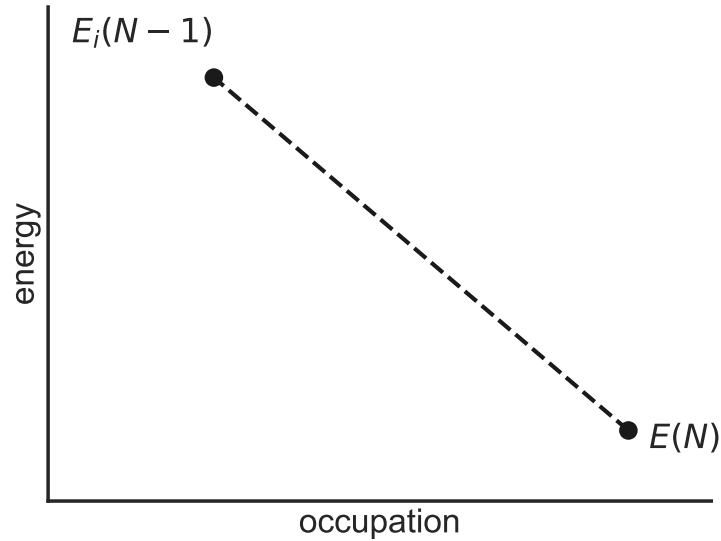




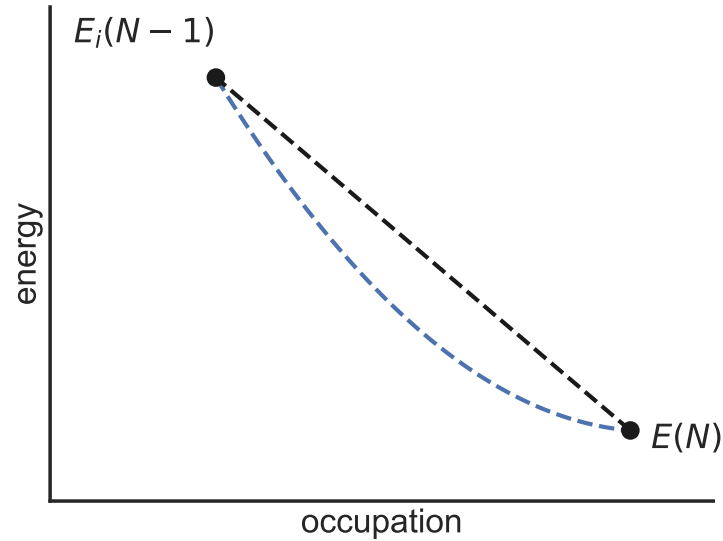
Screening: how to calculate the ideal α ?



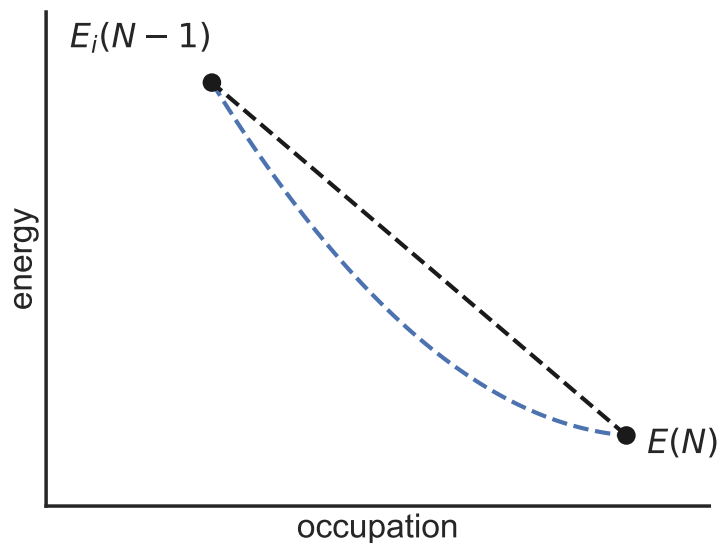
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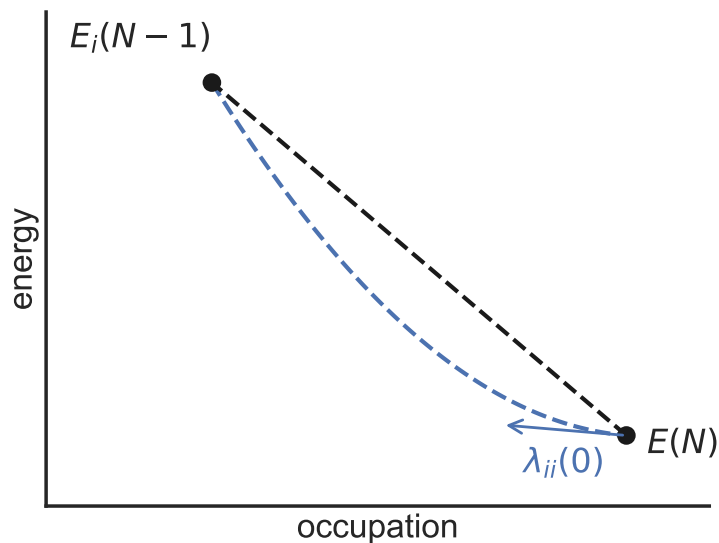


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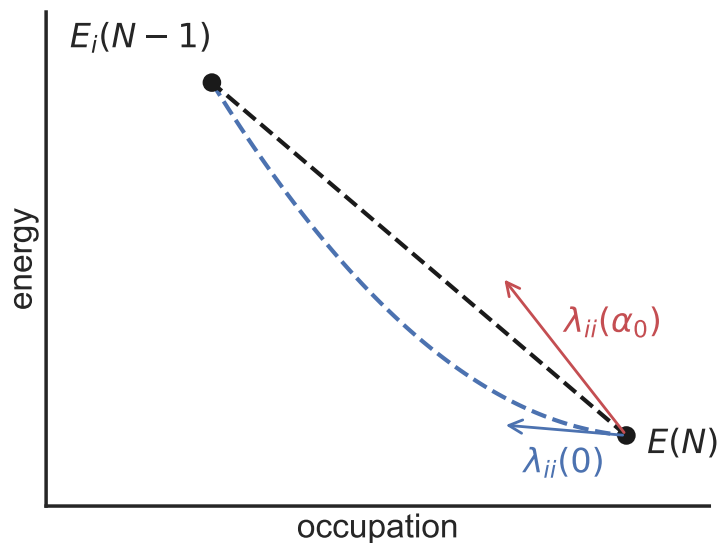
$$\lambda_{ii}(\alpha) \equiv \langle \varphi_i | \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} | \varphi_i \rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$

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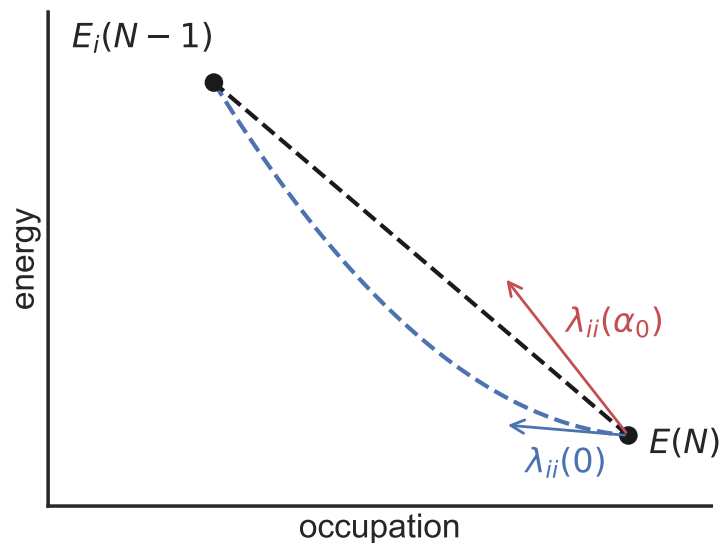
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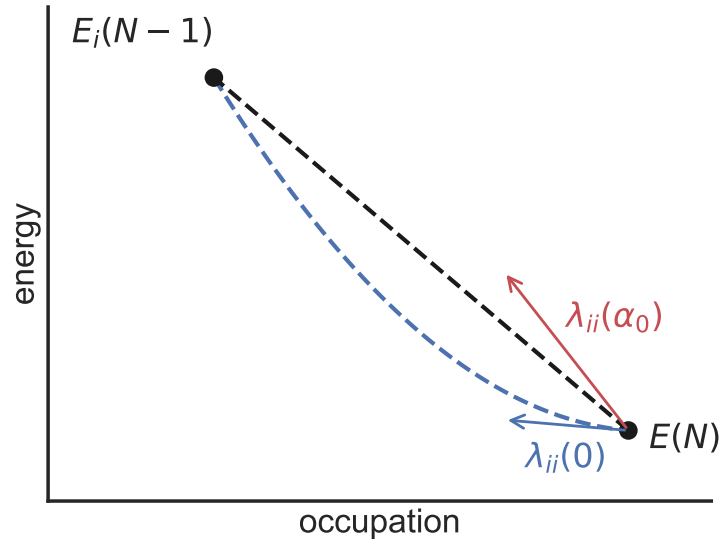
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Screening: how to calculate the ideal α ?



Given this, how to work out the ideal α ?

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$$\alpha_{n+1} = \alpha_n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha_n) - \lambda_{ii}(0)}$$

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trial screening
parameter

total energy with electron removed from orbital i

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Diagram illustrating the screening coefficient formula with annotations:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- α^n : trial screening parameter

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Diagram illustrating the iterative screening coefficient α^{n+1} calculation:

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- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: expectation value of $\hat{H}^{\text{Koopmans}}$ at $\alpha=0$
- $\lambda_{ii}(\alpha^n)$: expectation value of $\hat{H}^{\text{Koopmans}}$ at α^n
- α^n : trial screening parameter

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

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Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

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Reformulate in terms of DFPT...

$$\alpha_i = 1 + \frac{\langle v_{\text{pert}}^i | \Delta^i n \rangle}{\langle n_i | v_{\text{pert}}^i \rangle}$$

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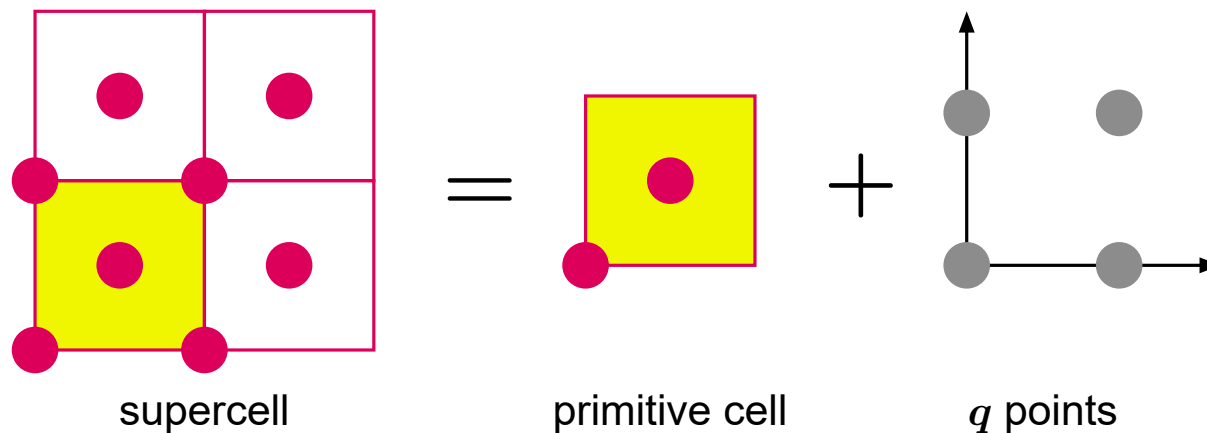
... in reciprocal space

$$\alpha_{\mathbf{0}i} = 1 + \frac{\sum_{\mathbf{q}} \langle v_{\text{pert},\mathbf{q}}^{\mathbf{0}i} | \Delta_{\mathbf{q}}^{\mathbf{0}i} n \rangle}{\sum_{\mathbf{q}} \langle n_{\mathbf{q}}^{\mathbf{0}i} | v_{\text{pert},\mathbf{q}}^{\mathbf{0}i} \rangle}$$

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Screening via DFPT



Orbital-density-dependence

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$$\begin{aligned} & - \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \\ & = E_{\text{Hxc}}[\rho] + E_{\text{Hxc}}[\rho - \rho_i] + f_i(-E_{\text{Hxc}}[\rho - \rho_i] + E_{\text{Hxc}}[\rho - \rho_i + n_i]) \end{aligned}$$

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For filled orbitals with KI:

$$v_i^{\text{KI}}/\alpha_i = -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}')$$

Orbital-density-dependence

Minimise $E[\rho, \rho_i]$ with CG minimisation:

G. Borghi *et al.* *Phys. Rev. B* **91**, 155112 (2015)

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outer loop $\varphi_i^{(n+1)} = \varphi_i^{(n)} + \Delta_i$

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For more details see Borghi *et al.* (2015)

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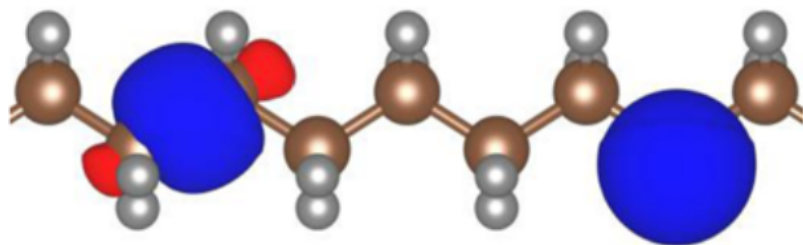
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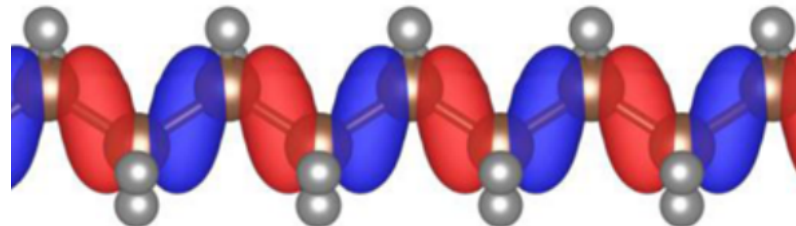
For more details see Borghi *et al.* (2015)

- gives rise to a set of **minimising orbitals** (localised/variational)
- diagonalising at the minimum gives rise to **diagonalising orbitals** (delocalised/canonical)

Orbital-density-dependence



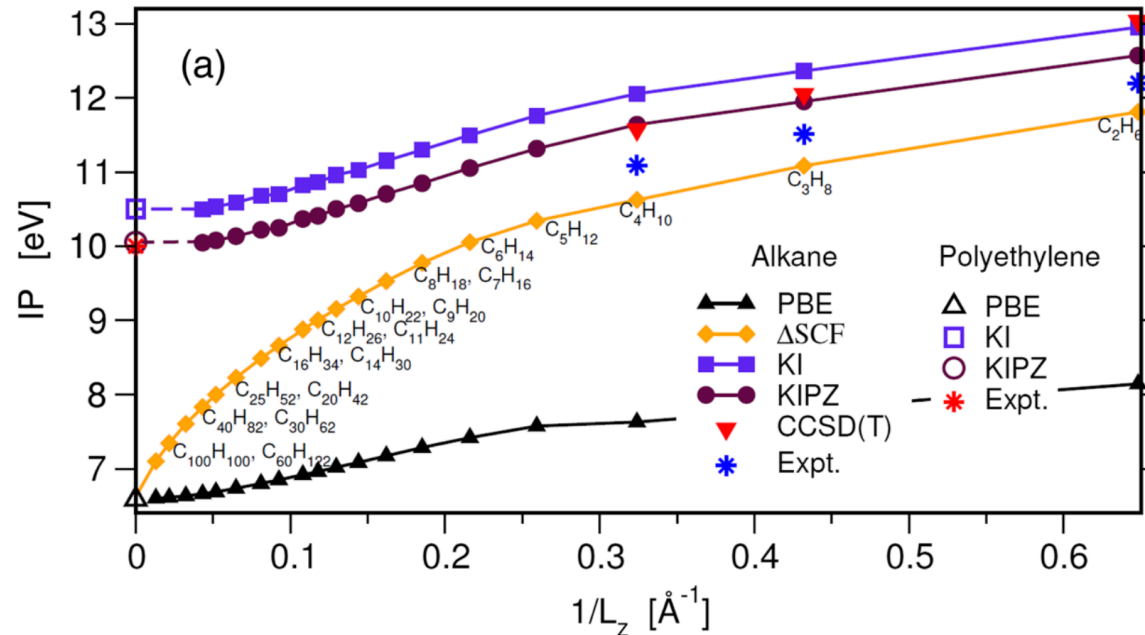
variational



canonical

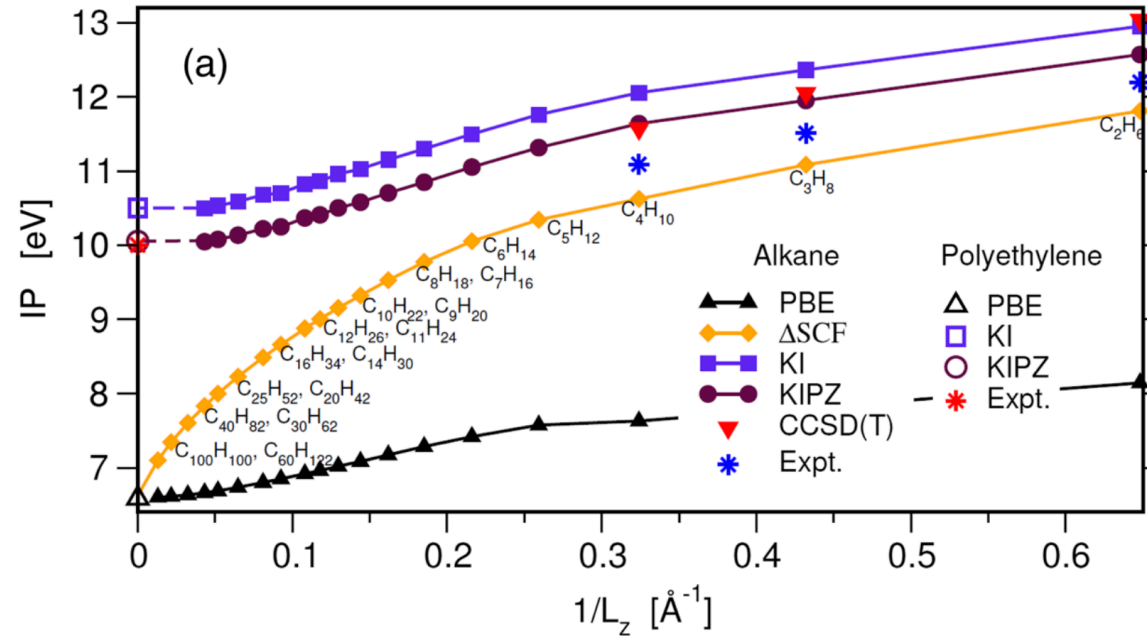
N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Importance of localisation



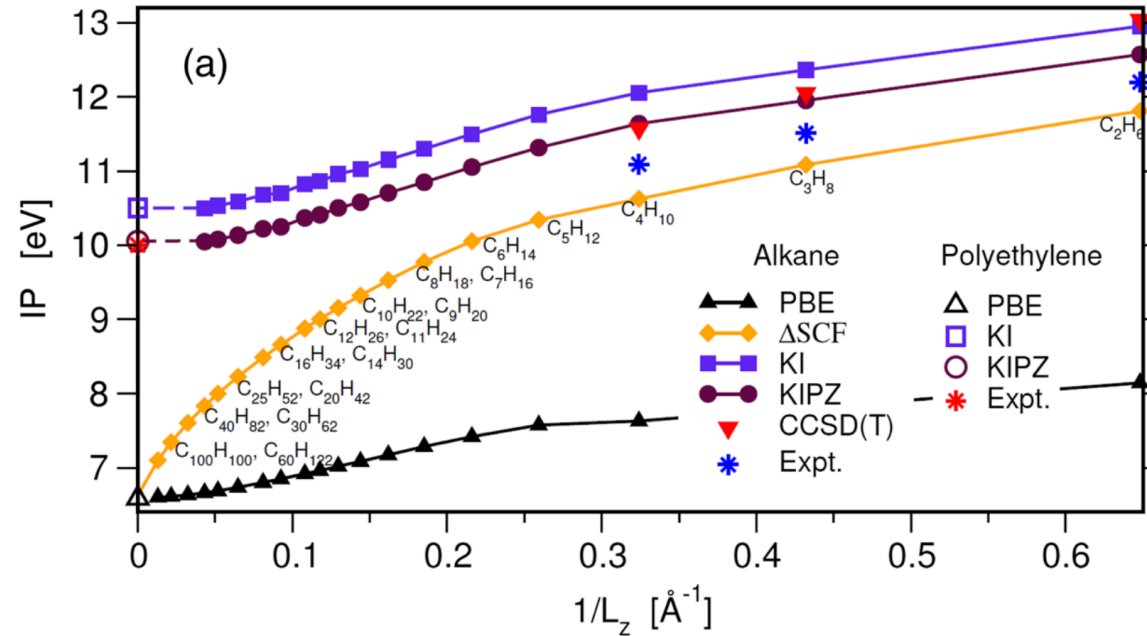
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Importance of localisation



In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

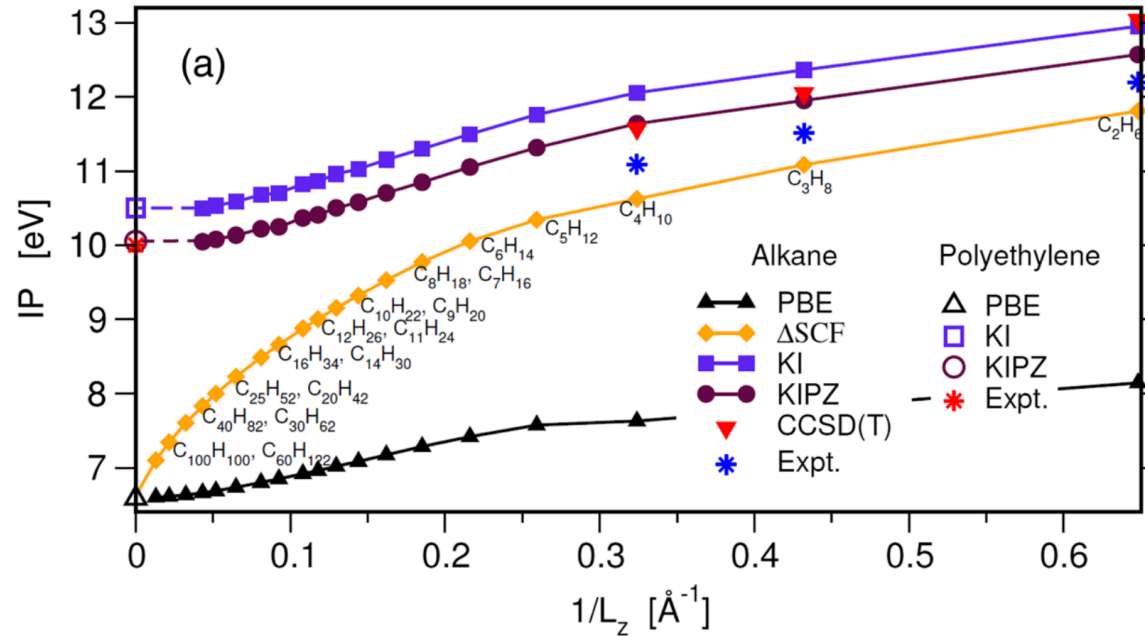
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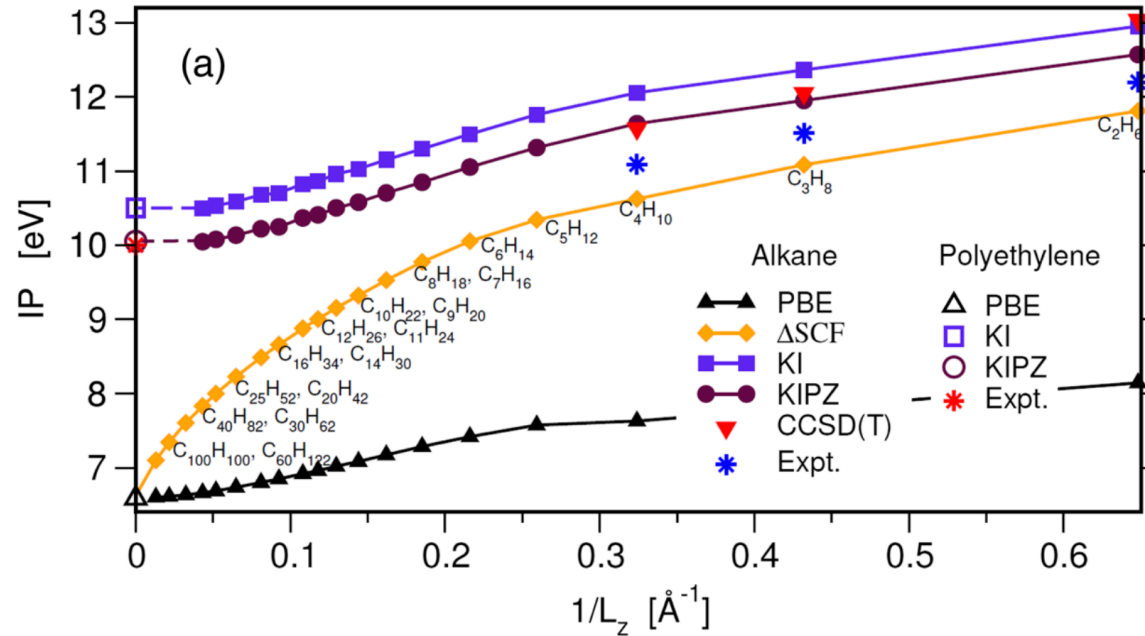
Across all the cells $\Delta E = \frac{1}{\delta N} (E(N - \delta N) - E(N)) = -\frac{dE}{dN} = -\varepsilon_{\text{HO}}$

Importance of localisation



$$\lim_{n_i(r) \rightarrow 0} v_i^{\text{KI}} / \alpha_i = \lim_{n_i(r) \rightarrow 0} -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}') = 0$$

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Solution: apply Koopmans correction to localised orbitals

Orbital-density-dependence

Other features of orbital-density-dependence

A. Ferretti *et al.* *Phys. Rev. B* **89**, 195134 (2014)

Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general

Orbital-density-dependence



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Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
- Practically we can often use MLWFs
- a natural generalisation in the direction of spectral functional theory

Flavours

Flavours of Koopmans functionals

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \eta_i \right)$$

One degree of freedom: what should be the gradient of this linear term?

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- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

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You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution

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- with a PZ correction $\rightarrow$ “KIPZ”

You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution
- “K” = an earlier iteration based off half-filling rather than integer endpoints (no longer used)

To summarise...

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When performing a Koopmans calculation, you must decide...

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- which flavour (KI, KIPZ)?

To summarise...

When performing a Koopmans calculation, you must decide...

- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?

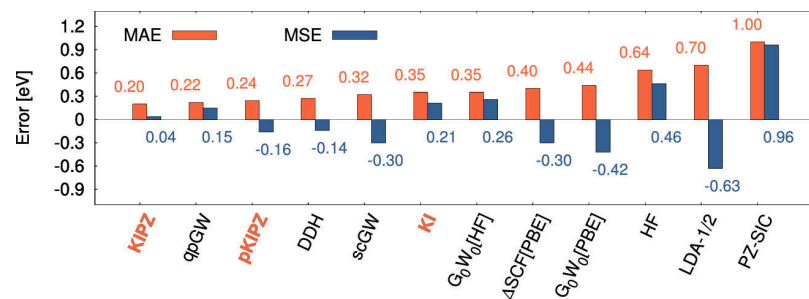
To summarise...

When performing a Koopmans calculation, you must decide...

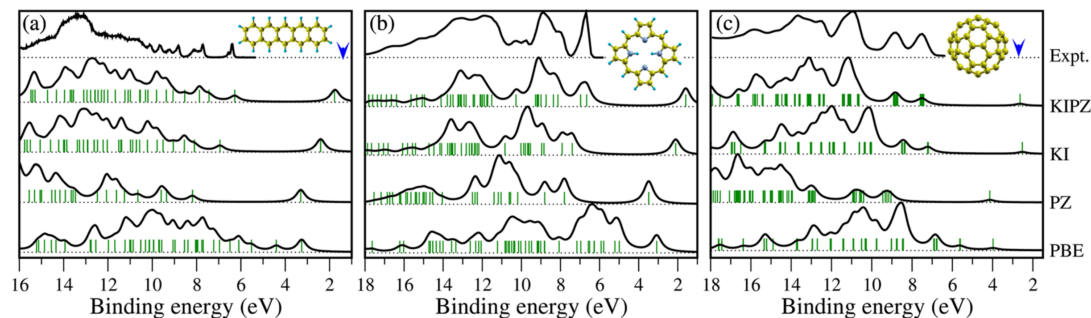
- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?
- how are we calculating the screening parameters? (finite differences, DFPT)

Results

Ionisation potentials of 100 molecules cf. CCSD(T)

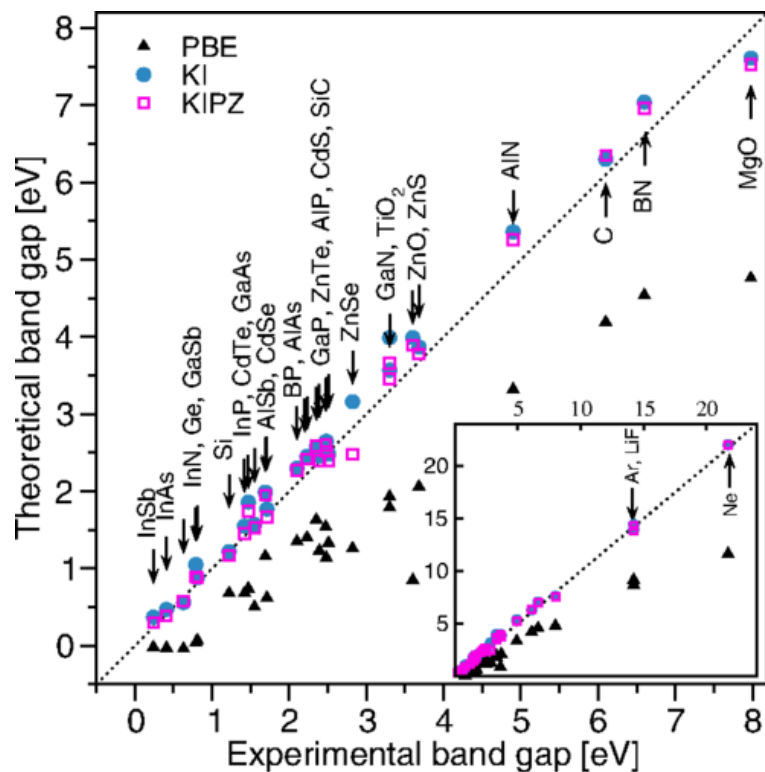


Ultraviolet photoemission spectra



N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018), N. L. Nguyen *et al.* *Phys. Rev. Lett.* **114**, 166405 (2015)

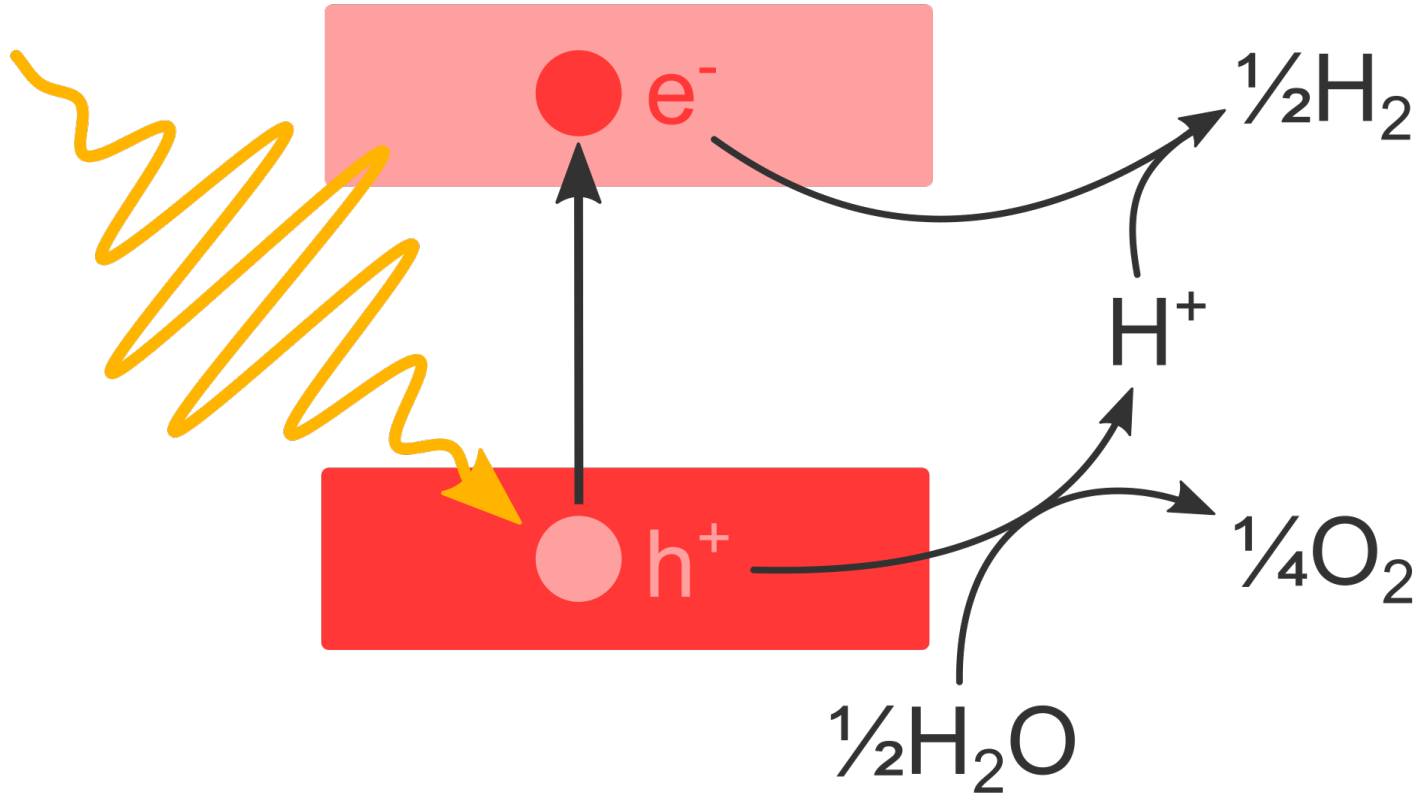
Koopmans functionals: results for solids



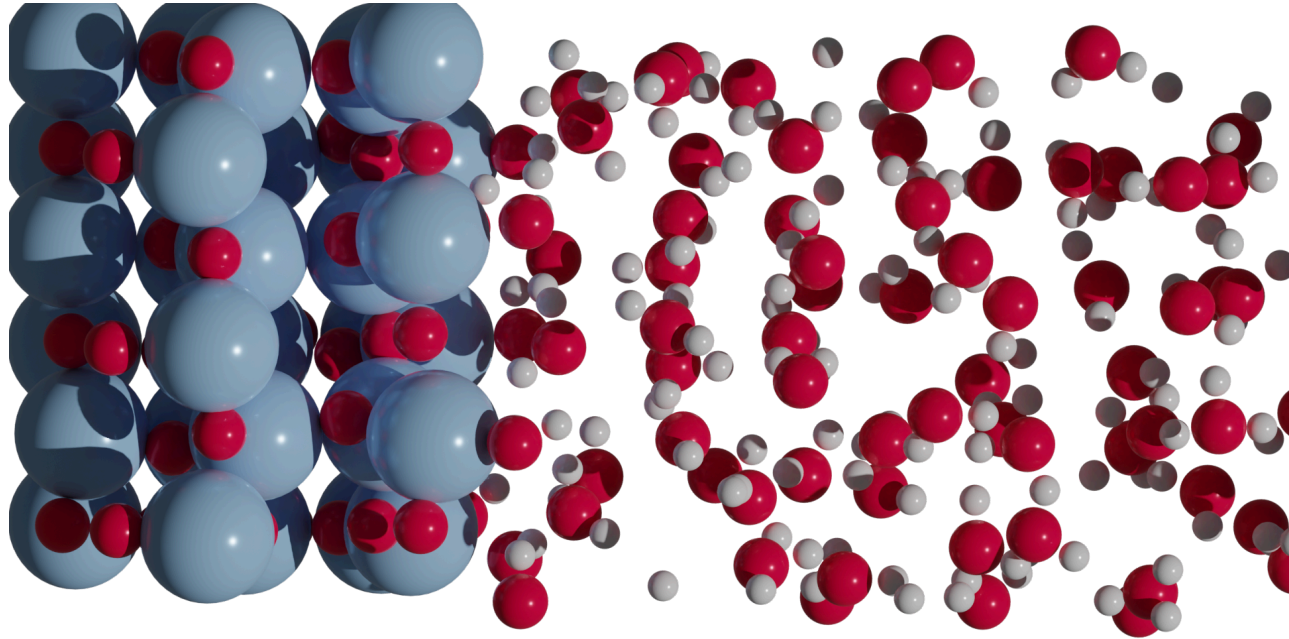
Mean absolute error (eV) across prototypical semiconductors and insulators

	PBE	G_0W_0	KI	KIPZ	QSGW $\tilde{W}$
E_{gap}	2.54	0.56	0.27	0.22	0.18
IP	1.09	0.39	0.19	0.21	0.49

Photocatalytic water-splitting



Photocatalytic water-splitting



M. Stojkovic *et al.* *Phys. Rev. Mater.* **10**, 25403 (2026)

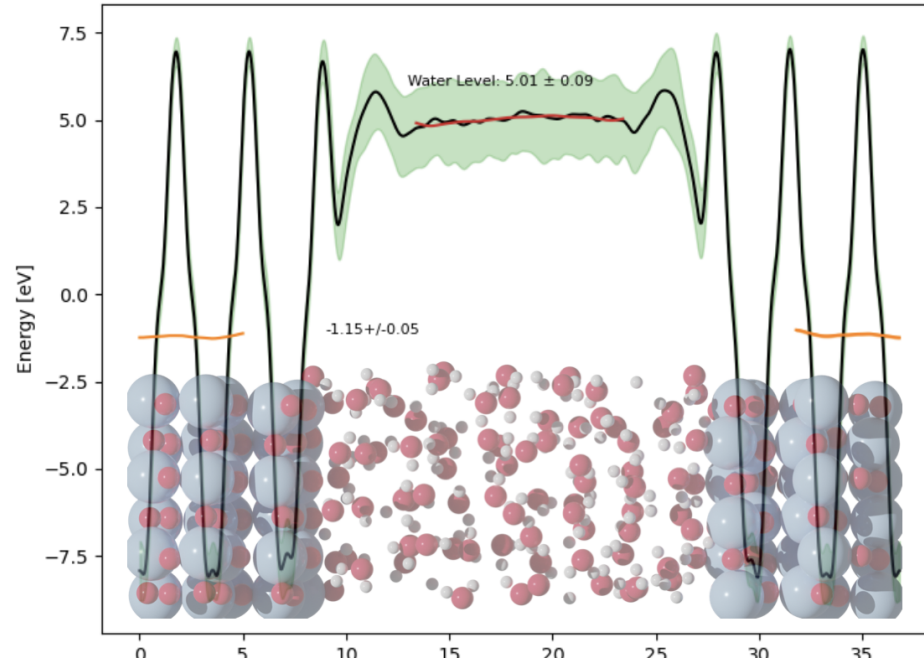
Photocatalytic water-splitting



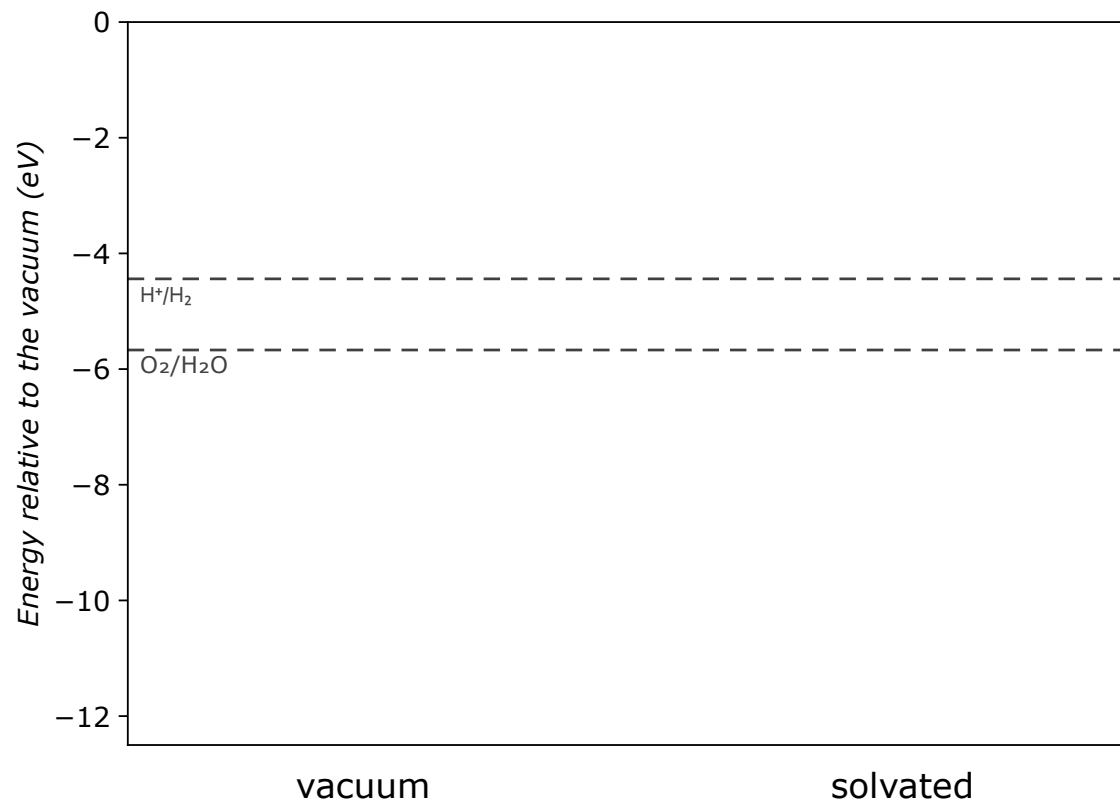
$$\varepsilon_i^{\text{aligned}} = \left(\varepsilon_i - \langle V_H \rangle_{\text{TiO}_2} \right)_{\text{bulk TiO}_2} + \left(\langle V_H \rangle_{\text{TiO}_2} - \langle V_H \rangle_{\text{water}} \right)_{\text{TiO}_2/\text{water}} + \left(\langle V_H \rangle_{\text{water}} - V_{\text{vac}} \right)_{\text{water/vac}}$$

PSI

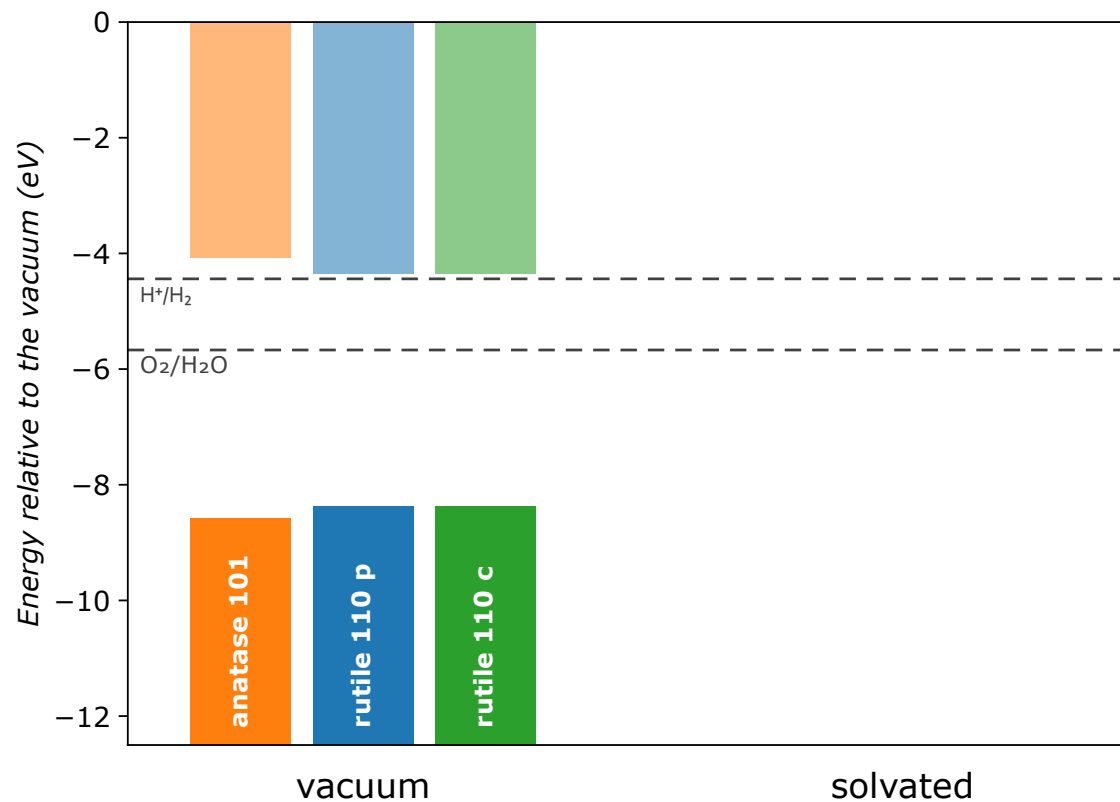
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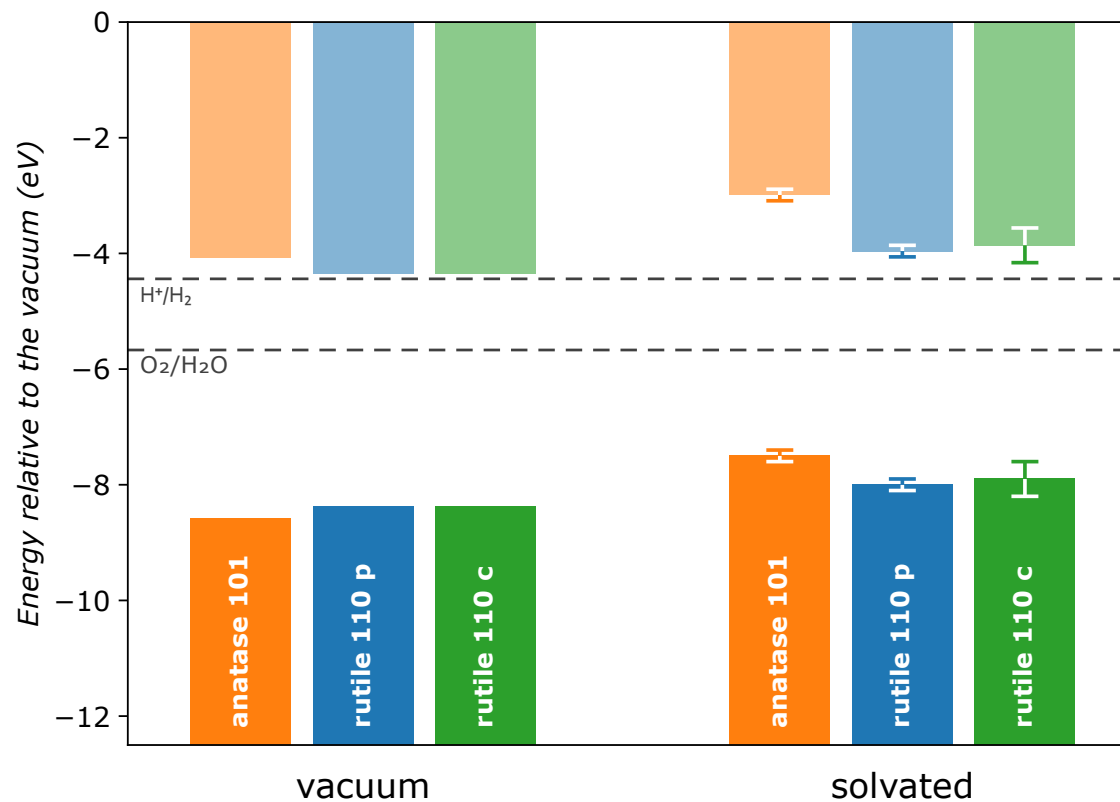
Band alignment for water-splitting



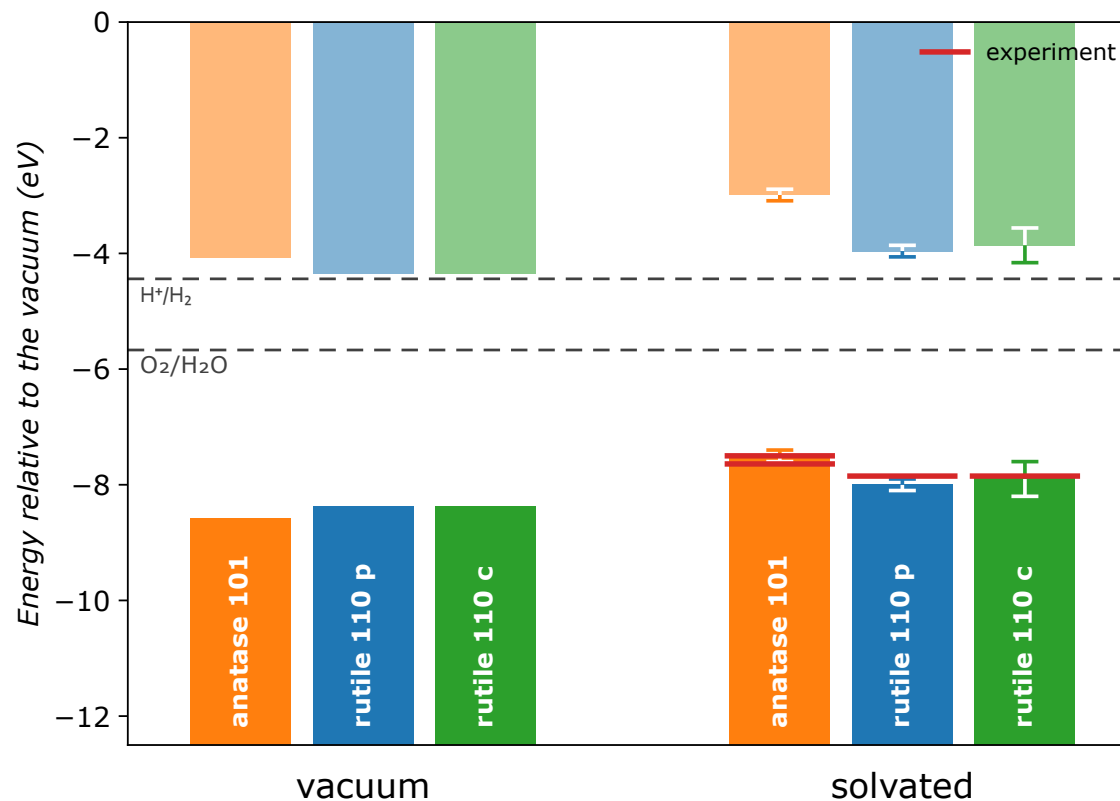
Band alignment for water-splitting



Band alignment for water-splitting

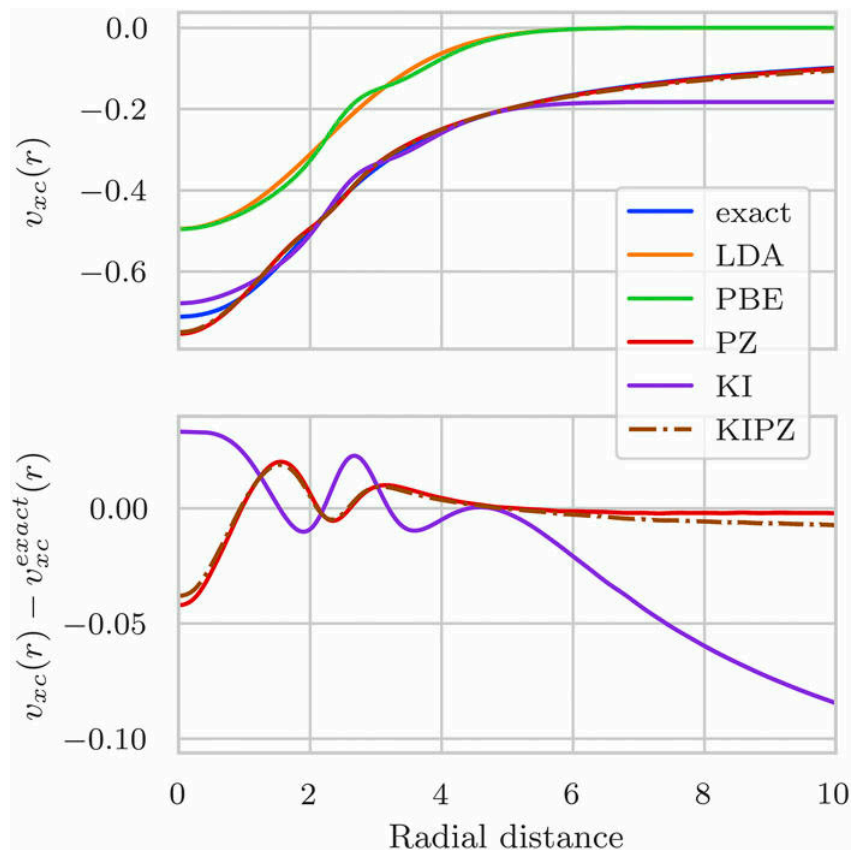


Band alignment for water-splitting



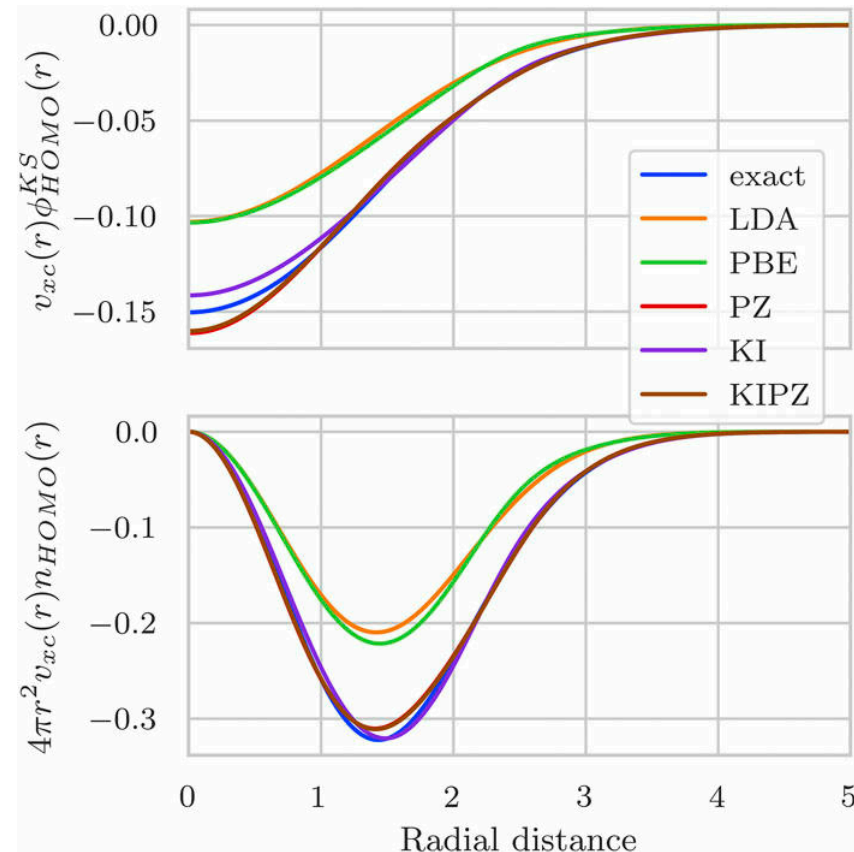
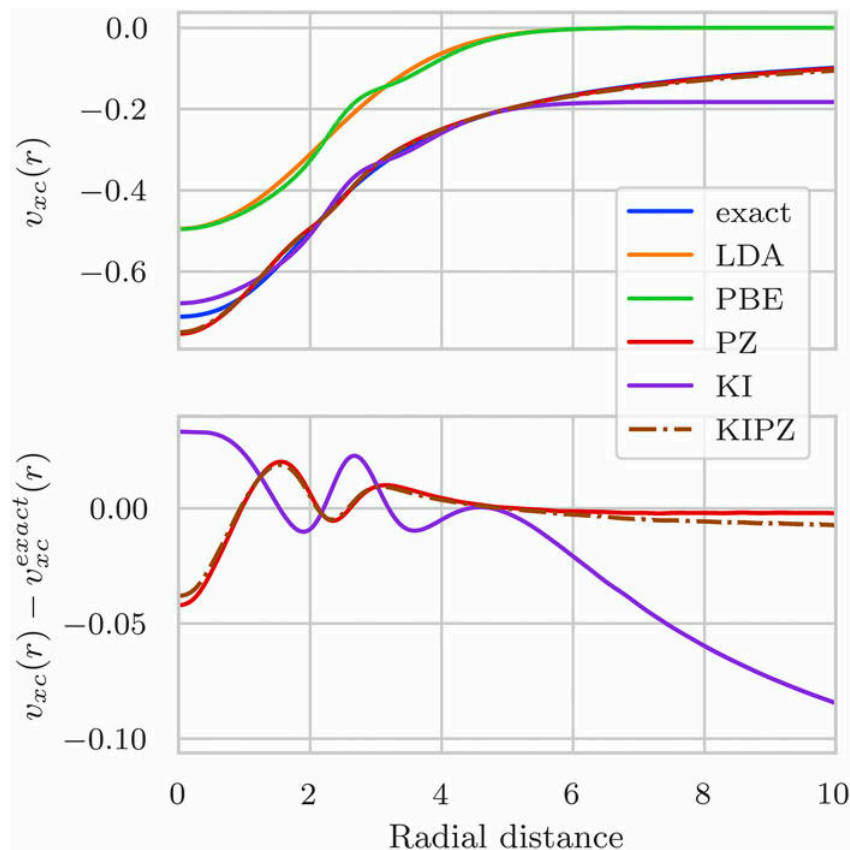
Toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



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Koopmans functionals: caveats

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- restricted to systems with a non-zero band gap

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- empty state localisation in the bulk limit

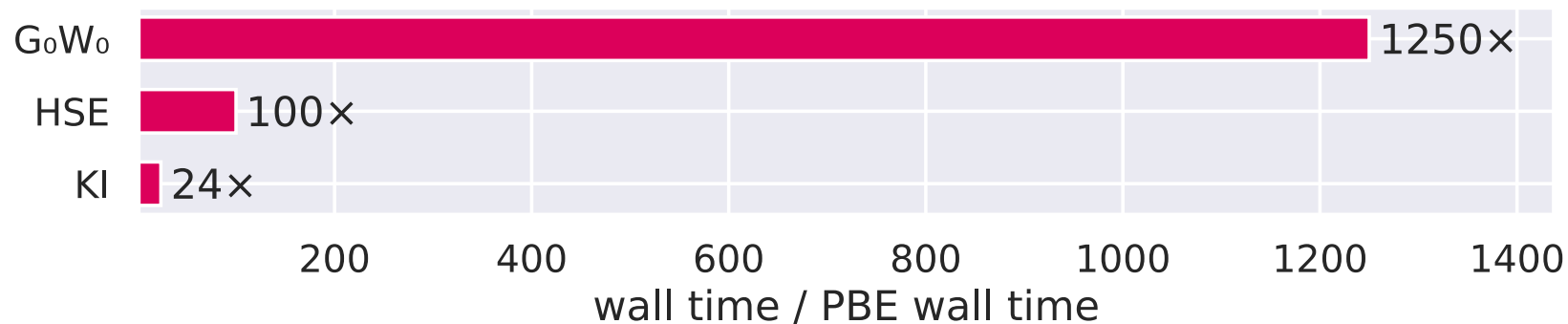
Koopmans functionals: caveats



- restricted to systems with a non-zero band gap
- empty state localisation in the bulk limit
- can potentially break the crystal point group symmetry

Computational cost and scaling

Computational cost and scaling



Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

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- one screening parameter per orbital

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

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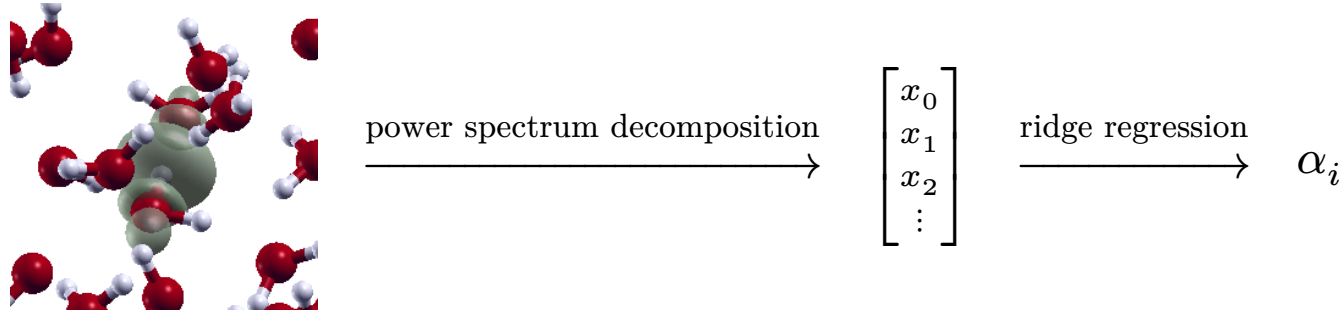
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 - DFPT: $\mathcal{O}(N_{\mathbf{k}}^2 N^3)$

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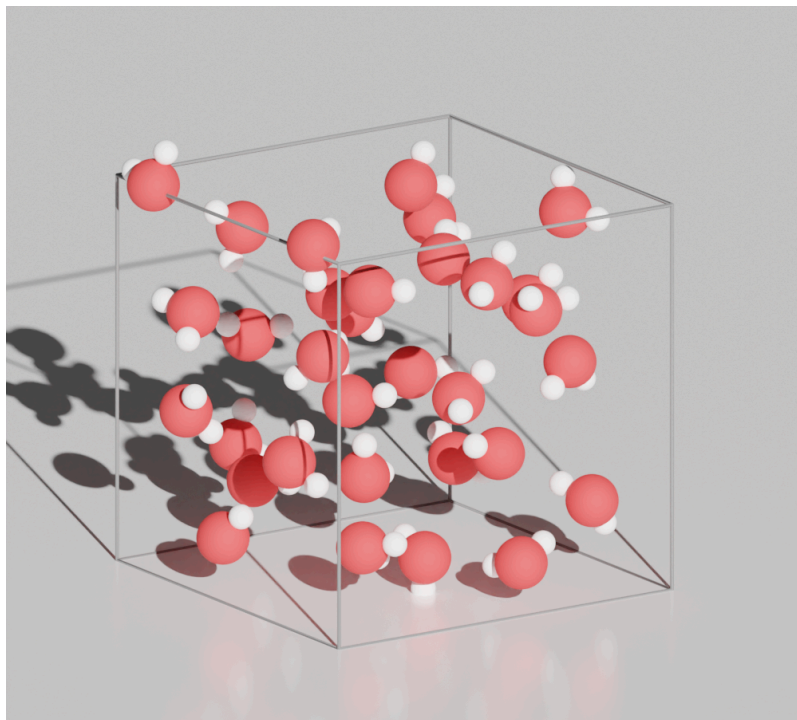
Machine-learned electronic screening



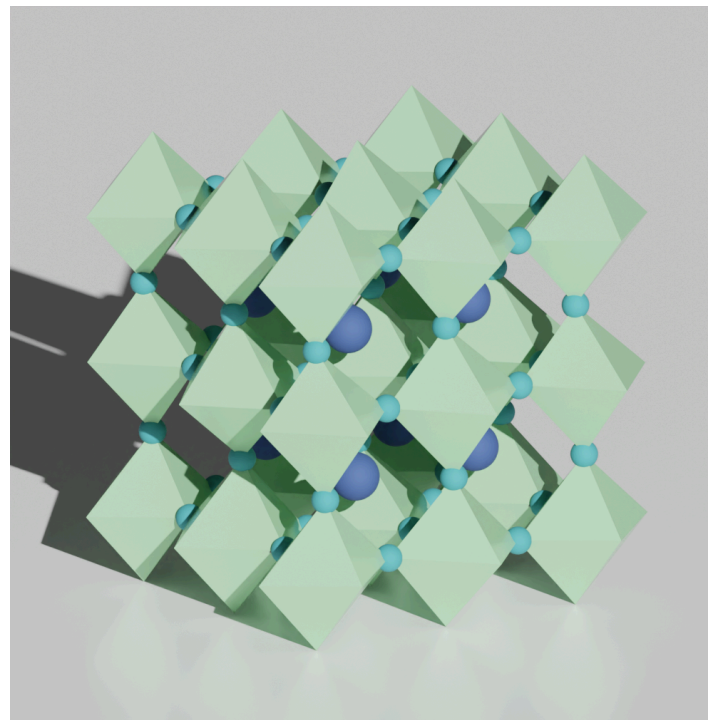
$$c_{nlm,k}^i = \int d\mathbf{r} g_{nl}(r) Y_{lm}(\theta, \varphi) n^i(\mathbf{r} - \mathbf{R}^i)$$

$$p_{n_1 n_2 l, k_1 k_2}^i = \pi \sqrt{\frac{8}{2l+1}} \sum_m c_{n_1 l m, k_1}^{i*} c_{n_2 l m, k_2}^i$$

Machine-learned electronic screening



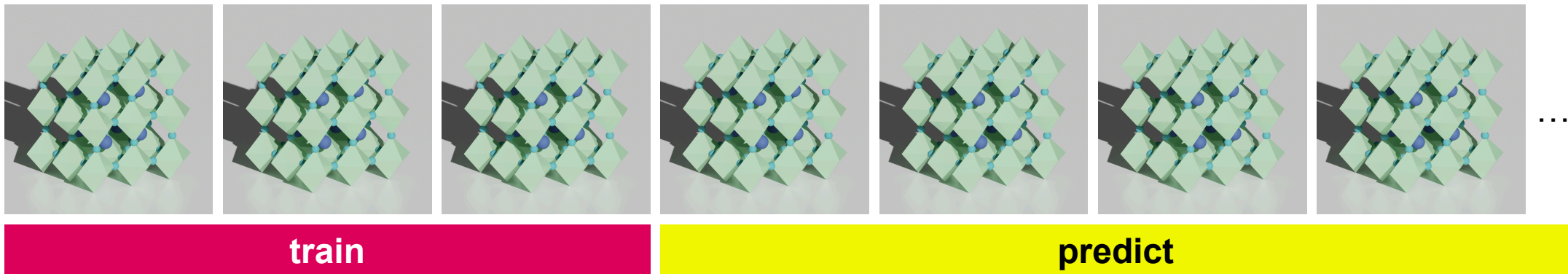
water



CsSnI_3

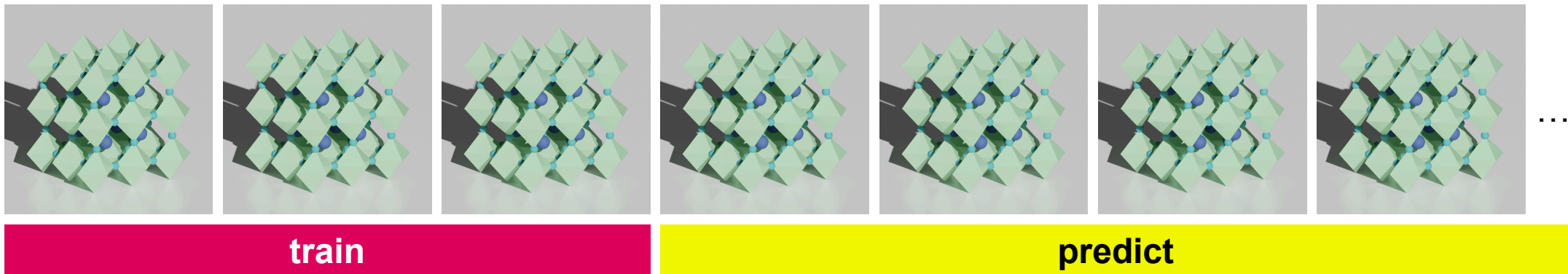
Machine-learned electronic screening

The use-case



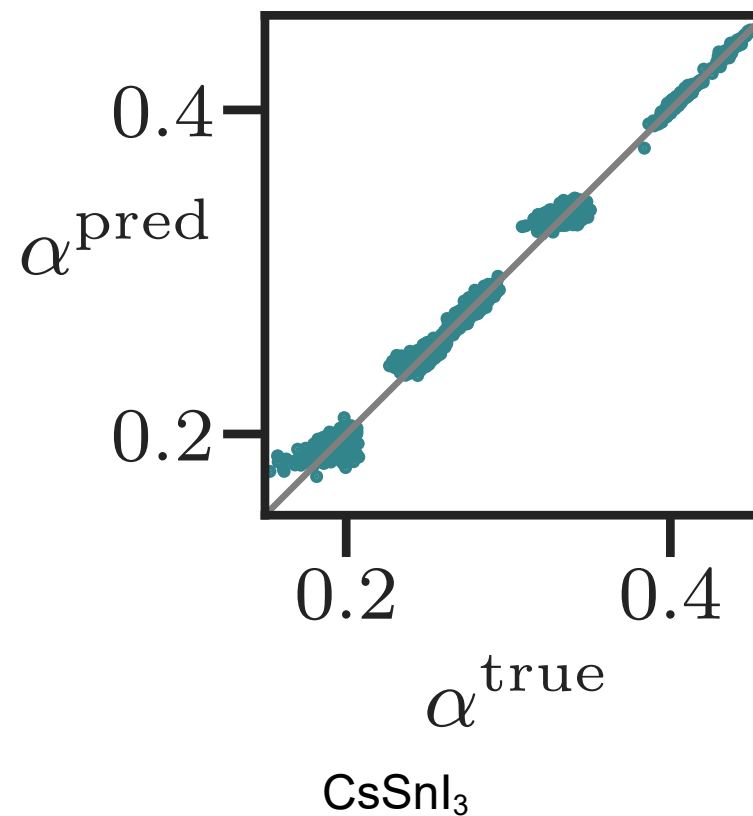
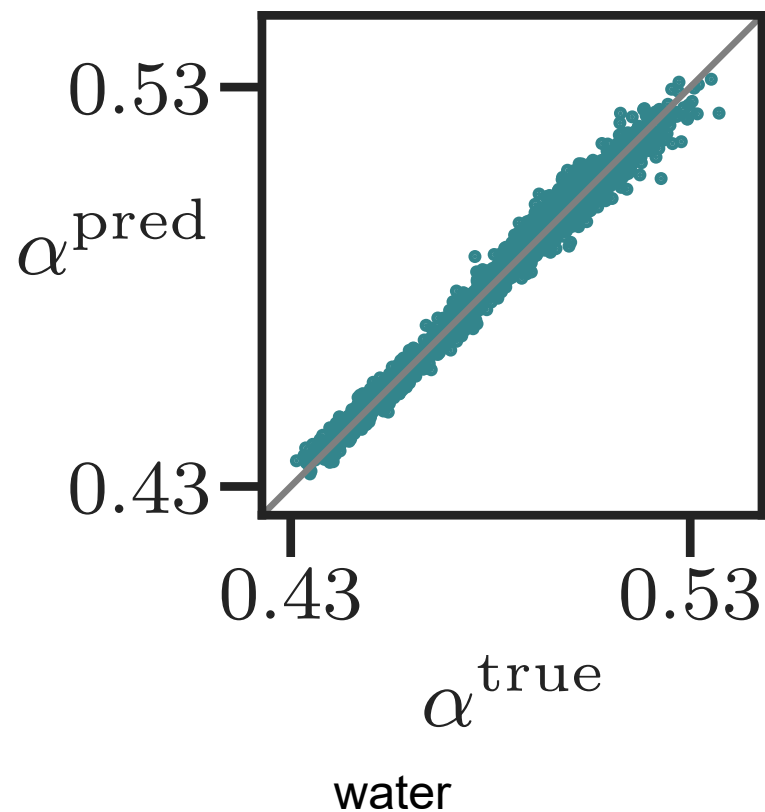
Machine-learned electronic screening

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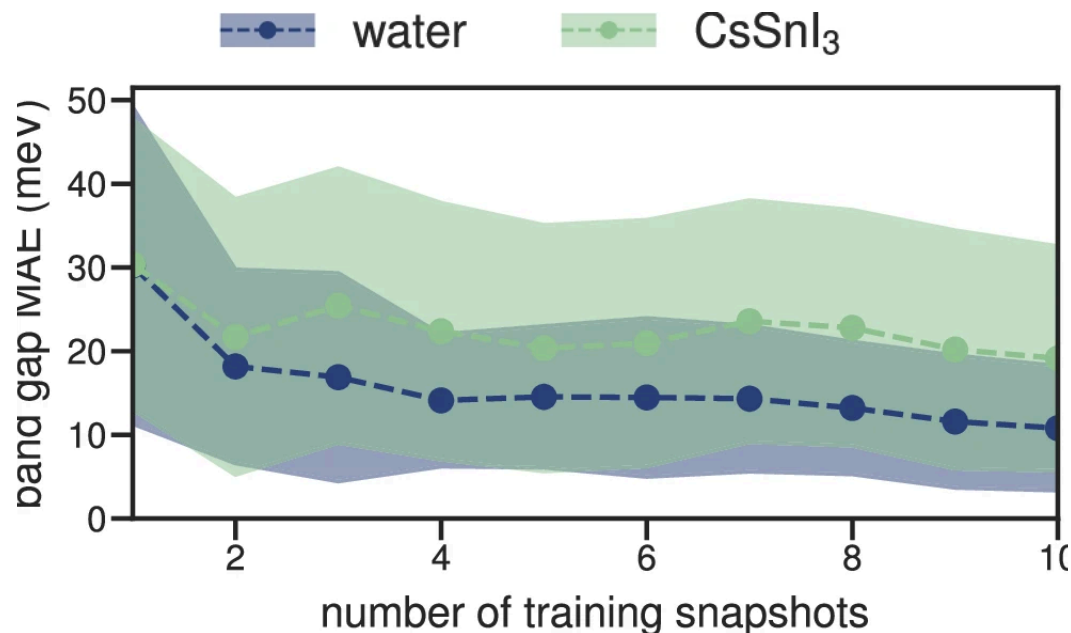


N.B. not a general model

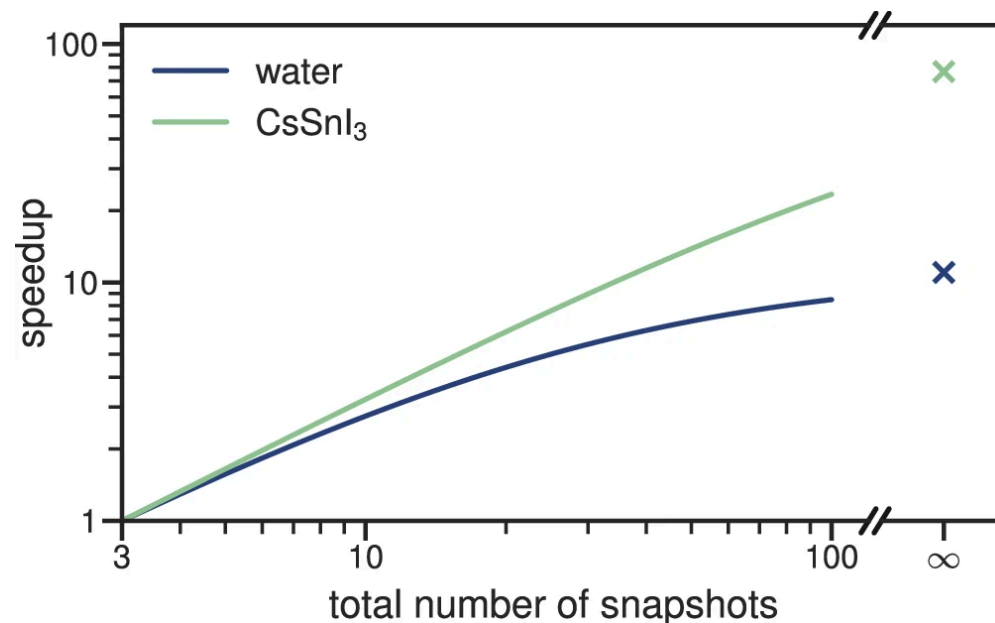
Machine-learned electronic screening



Machine-learned electronic screening



accurate to within $\mathcal{O}(10 \text{ meV})$ cf. typical band gap accuracy of $\mathcal{O}(100 \text{ meV})$



speedup of $\mathcal{O}(10)$ to $\mathcal{O}(100)$

Running a Koopmans functional calculation

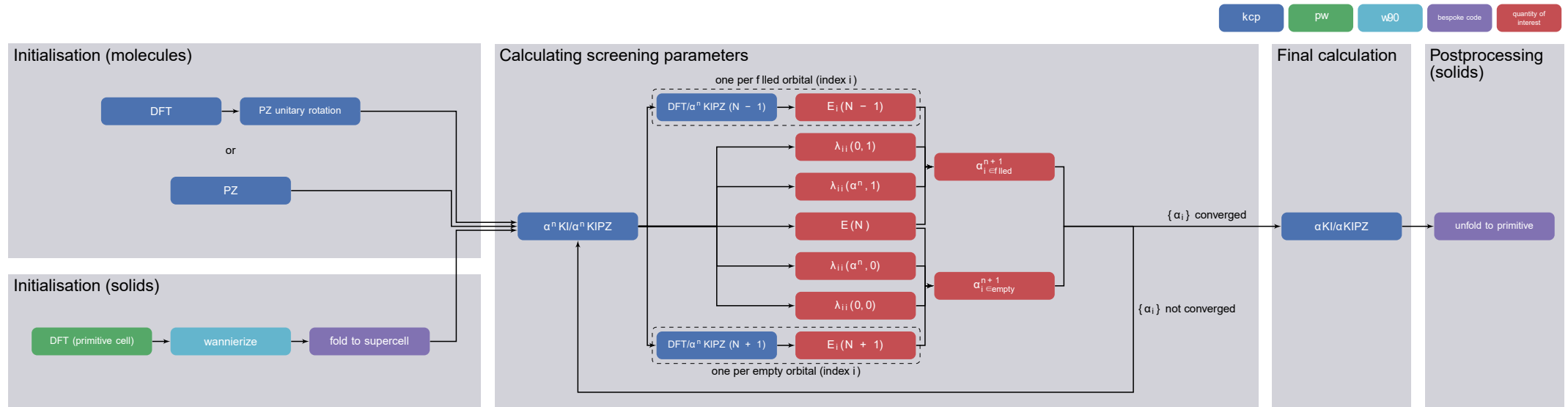
The workflows

The general workflow:

- define/initialise a set of variational orbitals
- calculate the screening parameters $\{\alpha_i\}$
- construct and diagonalise the Hamiltonian

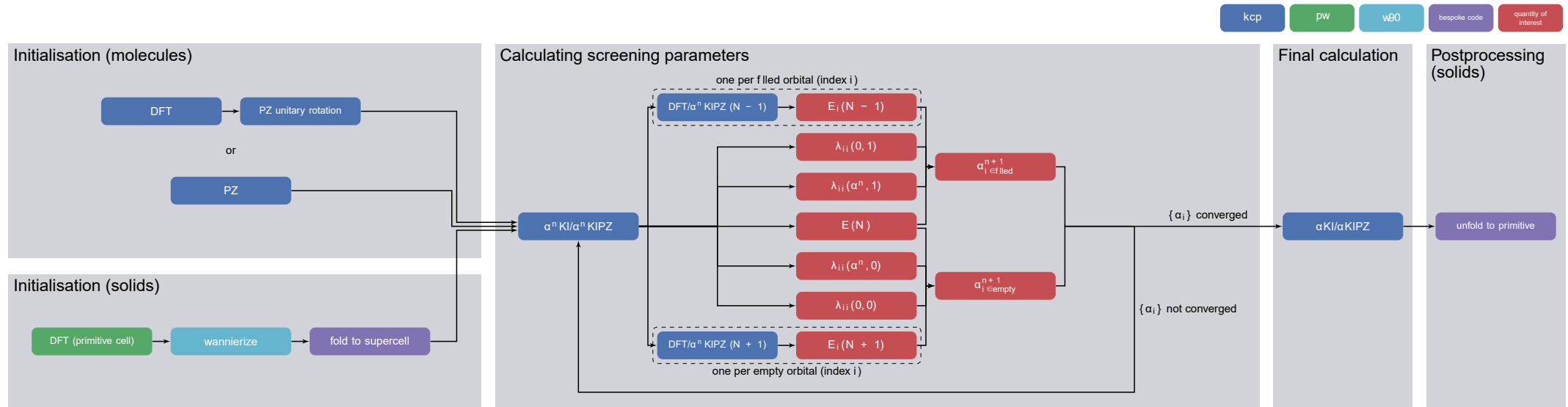
The workflows

(a) finite difference calculations using a supercell

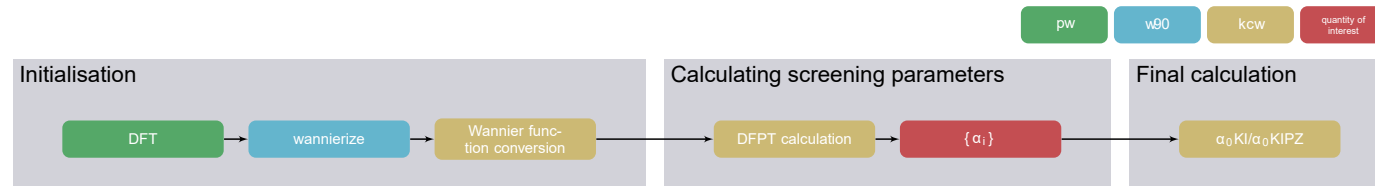


The workflows

(a) finite difference calculations using a supercell



(b) DFPT in a primitive cell

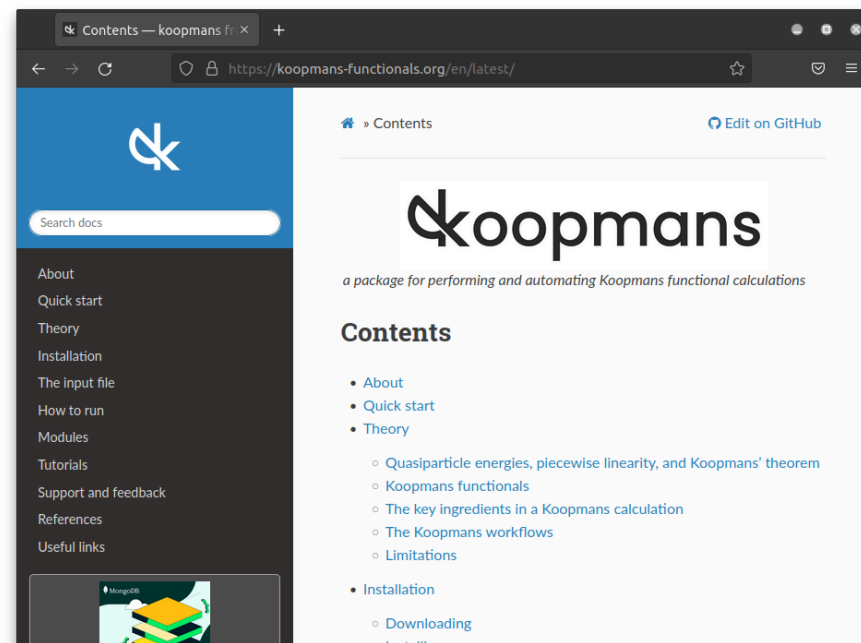


koopmans

The koopmans package

- v1.0 released in 2023 (now v1.2)
- implementations of Koopmans functionals
- automated workflows
 - start-to-finish Koopmans calculations
 - Wannierisation
 - dielectric tensor
 - ...
- built on top of ASE
- under the hood, calls QUANTUM ESPRESSO
- does not require expert knowledge

koopmans-functionals.org



E. B. Linscott *et al.* *J. Chem. Theory Comput.* **19**, 7097–7111 (2023)

A. H. Larsen *et al.* *J. Phys. Condens. Matter* **29**, 273002 (2017)

koopmans: the input file

```
{
  "workflow": {
    "task": "singlepoint",
    "functional": "ki",
    "method": "dscf",
    "init_orbitals": "mlwfs",
    "alpha_guess": 0.1
  },
  "atoms": {
    "atomic_positions": {
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      "positions": [
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        ["Si", 0.25, 0.25, 0.25]
      ]
    },
    "cell_parameters": {
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      "ibrav": 2,
      "celldm(1)": 10.262
    }
  },
}
```

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  "k_points": {
    "grid": [8, 8, 8],
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  },
  "calculator_parameters": {
    "ecutwfc": 60.0,
    "w90": {
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          "ang_mtm": "sp3"}],
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}]
      ],
      "dis_froz_max": 11.5,
      "dis_win_max": 17.0
    }
  }
}
```

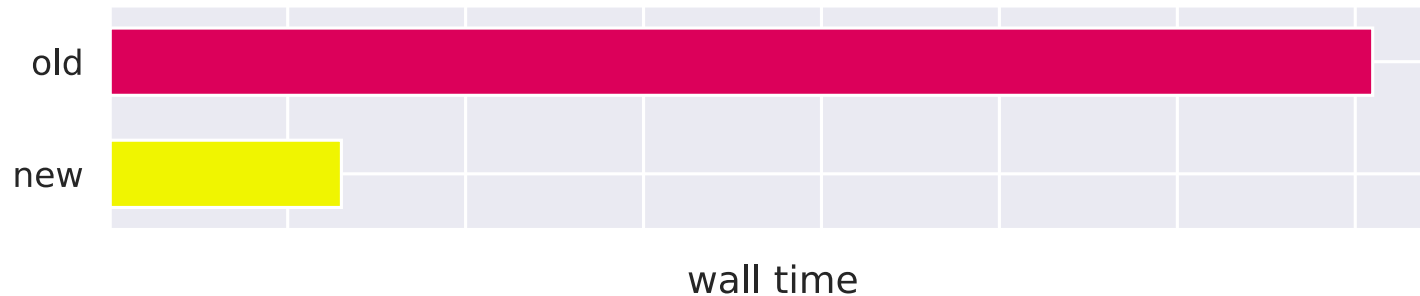
 **koopmans v2 is coming...** 

Koopmans AiiDA

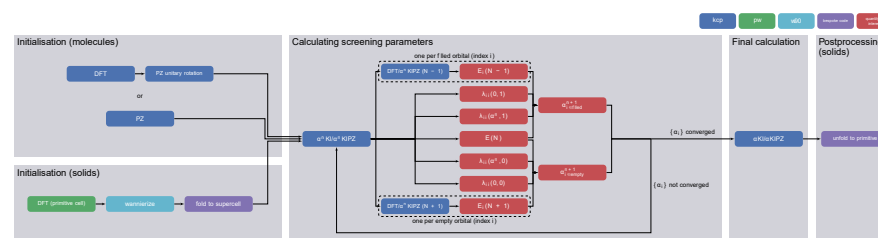
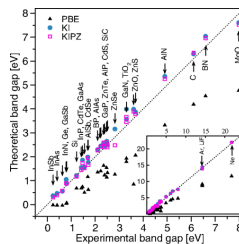
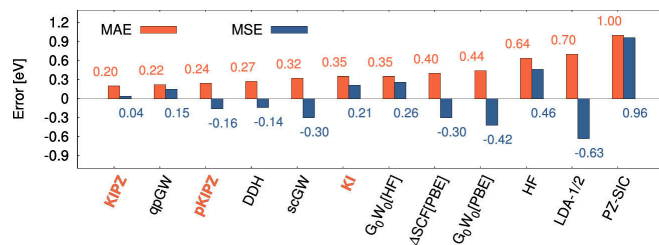
- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!

AiiDA

- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!



Take home messages



- Koopmans functionals are more complicated than a simple semi-local DFT calculation, because of...
 - orbital-density-dependence
 - screening parameters
- Koopmans functionals are implemented in QUANTUM ESPRESSO
- the complexity of the workflows are handled by the koopmans package
- keep your eyes peeled for v2!

koopmans: An Open-Source Package for Accurately and Efficiently Predicting Spectral Properties with Koopmans Functionals

Edward B. Linscott,^{*,△} Nicola Colonna,[△] Riccardo De Gennaro, Ngoc Linh Nguyen, Giovanni Borghi, Andrea Ferretti, Ismaila Dabo, and Nicola Marzari^{*}



Cite This: <https://doi.org/10.1021/acs.jctc.3c00652>



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Metrics & More

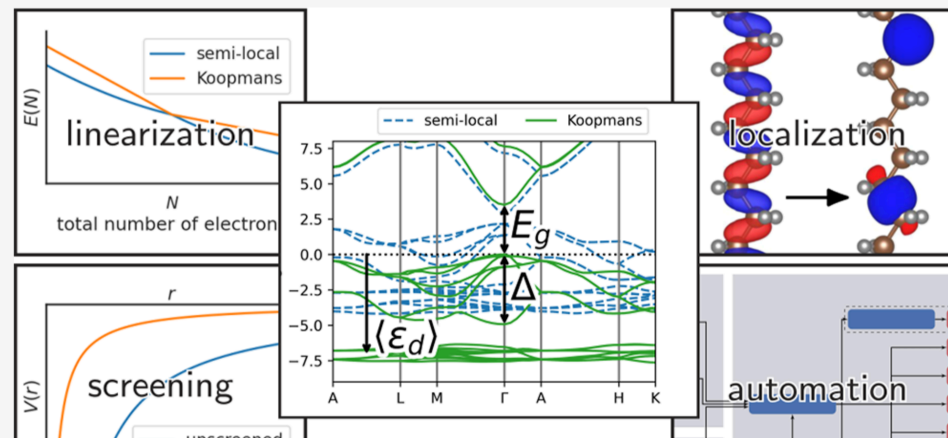


Article Recommendations



Supporting Information

ABSTRACT: Over the past decade we have developed Koopmans functionals, a computationally efficient approach for predicting spectral properties with an orbital-density-dependent functional framework. These functionals impose a generalized piecewise linearity condition to the entire electronic manifold, ensuring that orbital energies match the corresponding electron removal/addition energy differences (in contrast to semilocal DFT, where a mismatch between the two lies at the heart of the band gap problem and, more generally, the unreliability of Kohn–Sham orbital energies). This strategy has proven to be very powerful,



koopmans-functionals.org

Thank you!

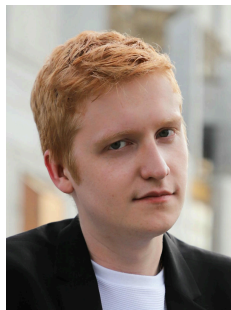
Acknowledgements



Nicola
Colonna



Miki
Bonacci



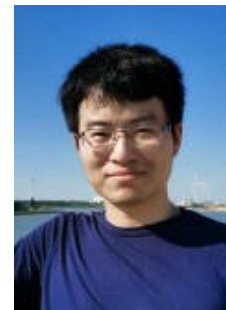
Aleksandr
Poliukhin



Marija
Stojkovic



Giovanni
Cistaro



Junfeng
Qiao



Yannick
Schubert



Nicola
Marzari



Want to find out more? Go to koopmans-functionals.org

slides available at  [github/elinscott-talks](https://github.com/elinscott-talks)

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- J. Qiao, G. Pizzi & N. Marzari. Automated mixing of maximally localized Wannier functions into target manifolds. *npj Comput Mater* **9**, 206 (2023).

Spare slides

Recap from earlier

Key idea: construct a functional such that the *variational* orbital energies

$$\varepsilon_i^{\text{Koopmans}} = \langle \varphi_i | H | \varphi_i \rangle = \partial E_{\text{Koopmans}} / \partial f_i$$

are...

- independent of the corresponding occupancies f_i
- equal to the corresponding total energy difference $E_i(N-1) - E(N)$

zero band gap $\rightarrow$ occupancy matrix for variational orbitals is off-diagonal

Automated Wannierisation

Koopmans functionals rely heavily on Wannier functions...

- to initialise the minimising orbitals, *or*
- in place of the minimising orbitals entirely

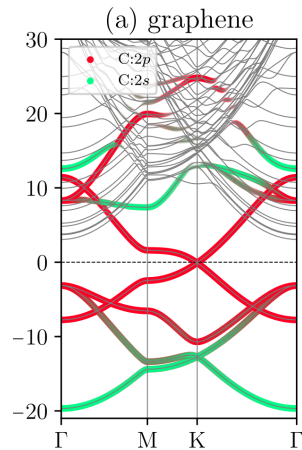
J. Qiao *et al.* *npj Comput Mater* **9**, 208 (2023)

J. Qiao *et al.* *npj Comput Mater* **9**, 206 (2023)

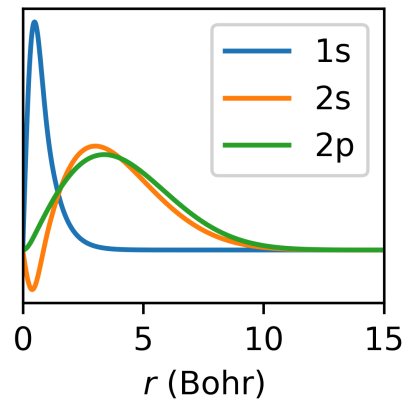
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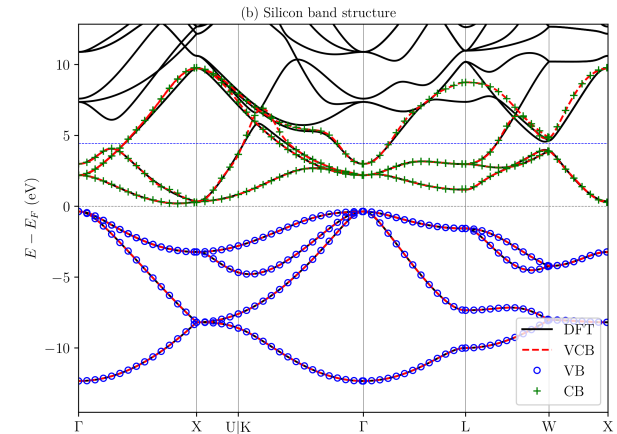
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projectability-based
disentanglement



use PAOs found in
pseudopotentials



parallel transport to separate manifolds